

**Title of Topic: Introduction to DBMS, Characteristics of DBMS,
DBMS vs File Systems, need for DBMS**

Course Code: 23IT201

Course Title: Database Management Systems

Session Number: 1

Academic Year: 2024 - 2025 (Even Sem)

Module 1 – INTRODUCTION

1. Introduction to DBMS, Characteristics of DBMS, DBMS vs File Systems, need for DBMS.
2. Three Level DBMS Architecture
3. Data Models Introduction, Benefits, and Phases
4. ER Diagrams – Symbols, Components, Relationships, Weak entities, Attributes, Cardinality.
5. Relational Algebra, Domain Relational Calculus, Tuple Relational
6. Calculus Keys - primary Key, Foreign Key.

Introduction to DBMS, Characteristics of DBMS, DBMS vs File Systems, need for DBMS

Topics to be Covered

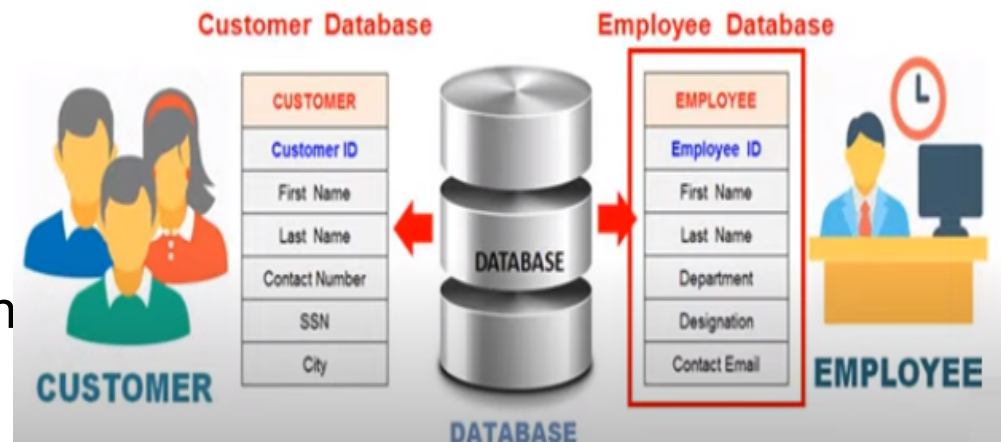
- DataBase (DB)
- DataBase Management System (DBMS)
- Applications of DBMS
- Metadata
- Very Large DataBase
- DBMS vs File System
- Users of DataBase System

DataBase (DB)

- Well organized collection of data that are related in a meaningful way which can be accessed in different logical orders but are stored only once
- The data (collection of raw facts and figures) in the database is therefore integrated, structured and shared
- Stores only persistent data (once stored it always stays there unless deleted by an explicit request)

Feature of Database

- Well organized and Related data
- Self-Describing
- Stored only once
- Accessible in different orders with



Database (DB)

DBMS stands for **D**atabase **M**anagement **S**ystem.

DBMS = Database + Management System.

Database is a collection of data and **Management System** is a set of programs to store and retrieve those data.

A **Database Management System (DBMS)** is a software system that is **designed to manage and organize data in a structured manner.**

It allows users to **create, modify, and query** a database, as well as **manage the security and access controls** for that database.

DataBase Management System (DBMS)

- Contains information about a particular enterprise
 - ✓ Collection of interrelated data
 - ✓ Set of programs to access the data
 - ✓ Environment that is both convenient and efficient to use
- Software system for creating, organizing and managing the database
- Provides an environment to perform operations on the database for creating, inserting, deleting, updating and retrieval of data



DBMS:



Database Management System (DBMS) – CHARACTERISTICS OF DBMS

1.Data modeling:

A DBMS provides **tools for creating and modifying data models**, which define the **structure and relationships** of the data in a database.

2.Data storage and retrieval:

A DBMS is responsible for **storing and retrieving data from the database**, and can provide **various methods for searching and querying the data**.

Database Management System (DBMS) –**CHARACTERISTICS OF DBMS****3.Concurrency control:**

A DBMS provides mechanisms for **controlling concurrent access to the database**, to ensure that **multiple users can access the data without conflicting with each other**.

4.Data integrity and security:

A DBMS provides tools for enforcing **data integrity and security constraints**, such as constraints on the values of data and access controls that **restrict who can access the data**

Database Management System (DBMS) –

CHARACTERISTICS OF DBMS

5. Backup and recovery:

A DBMS provides mechanisms for **backing up and recovering the data** in the **event of a system failure**.

6. DBMS can be classified into two types:

Relational Database Management System (RDBMS) and Non-Relational Database Management System (NoSQL or Non-SQL)

7. RDBMS:

Data is organized in the **form of tables** and each table has a set of **rows and columns**. The data is related to each other through **primary and foreign keys**.

Database Management System (DBMS) – CHARACTERISTICS OF DBMS

8.NoSQL:

Data is organized in the form of **key-value pairs**, **document**, **graph**, or **column-based**. These are designed to handle **large-scale**, **high-performance** scenarios.

Database Management System (DBMS) – CHARACTERISTICS OF DBMS



Applications of DBMS



- Online retailers: Order Tracking, Customized Recommendations
- Manufacturing: Production, Inventory, Orders, Supply Chain
- Human resources: Employee Records, Salaries, Tax Deductions

Metadata

- Data about data is called as Metadata
- Database includes information about how data is stored in computer storage

Document type	Book
Last name of author	Bloomfield
First name of author	Leonard
Year of publication	1933
Title	Language
City	New York
Publisher	Holt, Rinehart & Winston

Very Large DataBase

- Google, Yahoo, MSN - Search, Mail and Other Services
- Ebay - Buy and Sell
- YouTube - Videos (put & share)
- MySpace - Social Networking
- Facebook - Social Networking
- Wikipedia - Online Encyclopedia

Telecommunications

- Flickr - Share Photos
- Blogger - Own Blogs
- Fotolog - Share Photos & Blogs





DBMS vs File System

DBMS (Collection of Data)	File System (Collection of Data)
User is not required to write the procedures	User has to write the procedures for managing the database
Gives an abstract view of data that hides the details	Provides the detail of the data representation and storage of data
Provides a crash recovery mechanism, i.e., DBMS protects the user from the system failure	Doesn't have a crash mechanism, i.e., if the system crashes while entering some data, then the content of the file will lost
Provides a good protection mechanism	Very difficult to protect a file

DBMS vs File System

DBMS (Collection of Data)	File System (Collection of Data)
Contains a wide variety of sophisticated techniques to store and retrieve the data	Can't efficiently store and retrieve the data
Takes care of Concurrent access of data using some form of locking	Concurrent access has many problems like redirecting the file while other deleting some information or updating some information
Redundancy control is available in DBMS	Redundancy control is not available in file system
Unauthorized access is restricted in DBMS	Unauthorized access is not restricted in the file system

DBMS vs File System

DBMS (Collection of Data)	File System (Collection of Data)
Provide back up and recovery	Data lost in file system can't be recovered
Provide multiple user interfaces	Data is isolated in file system
Coordinates both the physical and the logical access to the data	Coordinates only the physical access
Designed to allow flexible access to data (i.e. queries)	Designed to allow predetermined access to data (i.e. compiled programs).
Designed to coordinate multiple users accessing the same data at the same time	Designed to allow one or more programs to access different data files at the same time

Users of DataBase System

DataBase Administrator (DBA)

- Manage all level of DBMS model
- DBA is responsible for
 - ✓ Schema definition
 - ✓ Storage structure and access-method definition
 - ✓ Schema and physical-organization modification
 - ✓ Granting of authorization for data access
 - ✓ Acquiring s/w and h/w resources as needed
 - ✓ Routine maintenance

End Users

- Access to the database for querying, updating, and generating reports

Users of DataBase System

DataBase Designers

- Logical Designers - Identifying the data to be stored in the database and relationship between them
- Physical Designers - Choosing appropriate structures to represent and store data

System Analysts and Application Programmers

- System Analysts communicate with all prospective database users, in order to understand their requirements especially naive and parametric end users, and develop specifications
- Application programmers implement specifications as programs; then test, debug, document, and maintain canned transactions

Users of DataBase System

Casual End Users

- Occasionally access the database
- Need different information each time
- Learn only a few facilities that they may use repeatedly

Naive or Parametric or Unsophisticated End Users

- Constantly querying and updating the database, using standard types of queries and updates
- Need to learn very little about the facilities provided by the DBMS
- Reservation clerks for airlines, hotels, and car rental companies check availability for a given request and make reservations

Users of DataBase System

Sophisticated End Users

- Engineers, scientists, business analysts, and others who thoroughly familiarize themselves with the facilities of the DBMS
- Try to learn most of the DBMS facilities in order to achieve their complex requirements

Stand-Alone Users

- Maintain personal databases by using ready-made program packages that provide easy- to-use menu or graphics-based interfaces
- An example is the user of a tax package that stores a variety of personal financial data for tax purposes

Users of DataBase System

Tool Developers

- Tools are optional packages that are often purchased separately
- Include packages for database design, performance monitoring, natural language or graphical interfaces, prototyping, simulation, and test data generation

Operators and Maintenance Personnel

- System administration personnel who are responsible for the actual running and maintenance of the hardware and software environment for the database system

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Summary

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Try Yourself

- A person has to create a database for a university so that the university will have all its data in the database. Identify the data to be added for the university database.
- Maintaining a customer database is an important way to keep up-to-date on customer information and data. An online shopping system maintains all its customer data in a database. List out the data available in such a customer database.
- An employer has to maintain his employee details in a database. He informs one of his employee to create a database to store the details of all the employees. Organize the data for the employee database.

Session 2 - DBMS Architecture, Data Models – Introduction, Benefits, and Phases